Quadratics: Expanding & Factorising

Practice test 4 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 - 4x - 9x + 36$ (1) = $\mathbf{x^2 - 13x + 36}$ (1)

A2. $15 + 5x - 3x - x^2$ (1) = $\mathbf{15 + 2x - x^2}$ (1)

A3. $9x^2 - 6xy - 6xy + 4y^2$ (1) = $\mathbf{9x^2 - 12xy + 4y^2}$ (1)

A4. $(2x^2 - 5x - 3) - (x^2 - 4x + 4)$ (1) = $\mathbf{x^2 - x - 7}$ (1)

Section B — Expanding three brackets

B1. $(x + 5)(x - 5) = x^2 - 25$ (1) $(x^2 - 25)(x + 2)$, with at least 3 of the 4 terms correct (1) = $\mathbf{x^3 + 2x^2 - 25x - 50}$ (1)

B2. $(x + 2)^2 = x^2 + 4x + 4$ (1) $(3x - 1)(x^2 + 4x + 4) = 3x^3 + 12x^2 + 12x - x^2 - 4x - 4$ (1) = $\mathbf{3x^3 + 11x^2 + 8x - 4}$ (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to 63, add to –16: –7 and –9 (1) **(x – 7)(x – 9)** (1)

C2. Multiply to –84, add to 5: 12 and –7 (1) **(x + 12)(x – 7)** (1)

C3. (2x + 1)(2x – 1) (1)

C4. $(x^2 + 9)(x^2 - 9)$ (1) = $\mathbf{(x^2 + 9)(x + 3)(x - 3)}$ (1)

Section D — Factorising $ax^2 + bx + c$

D1. $6 \times -2 = -12$; numbers: 4 and –3, so $6x^2 + 4x - 3x - 2 = 2x(3x + 2) - 1(3x + 2)$ (1) **(2x – 1)(3x + 2)** (1)

D2. $10 \times 3 = 30$; numbers: –15 and –2, so $10x^2 - 15x - 2x + 3 = 5x(2x - 3) - 1(2x - 3)$ (1) **(5x – 1)(2x – 3)** (1)

D3. $2(4x^2 - x - 3)$ (1) = **2(4x + 3)(x – 1)** (1)

Section E — Using factorising

E1. $3x^2 - 5x - 2 = 0$ (1) $(3x + 1)(x - 2) = 0$ (1) $\mathbf{x = -\frac{1}{3}}$ **or x = 2** (1)

E2. Numerator = $(2x + 3)(2x - 3)$ (1) denominator = $(2x - 3)(x + 2)$ (1) cancel $(2x - 3)$: **(2x + 3) / (x + 2)** (1)

Section F — Problem solving & proof

F1. $2(x^2 + 6x) + 7$ (1) = $2[(x + 3)^2 - 9] + 7$ (1) = $\mathbf{2(x + 3)^2 - 11}$ (1)

F2. $(4n^2 + 4n + 1)$ (1) – $(4n^2 - 4n + 1)$ (1) = 8n, which is **8 × an integer, so a multiple of 8** (1)

F3. $(8.7 + 1.3)(8.7 - 1.3) = 10 \times 7.4 =$ **74** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: $(x – 7)(x + 4)$ is the same as $(x + 4)(x – 7)$. “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.